

Ariel Campêlo Viana Morais

Backend Developer / Artificial Intelligence & Software Engineer

Contact Information

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Professional Summary

B.Sc. in **Mathematics from the University of São Paulo (USP)** specializing in **Symbolic Artificial Intelligence (Formal Logic & Ontologies)**, **Natural Language Processing (NLP)**, and **Backend Software Engineering**. Researcher at the **Institute of Software Technology (ITS)** and Consultant at **Fundação Getúlio Vargas (FGV/IBRE)**, leading AI system architecture for international electronic sanitary certificate exchanges (**e-CERT / UN/CEFACT / MAPA**), agribusiness data harmonization (CONSECANA-SP), and semantic platforms. Creator of a personal ecosystem of intelligent web apps (PWAs) designed to optimize daily productivity and personal well-being.

Key Competencies & Technical Stack

- **Artificial Intelligence & Semantics:** Symbolic AI, Ontologies (OWL, RDF, SPARQL), SNOMED CT, ICD, UN/CEFACT Standards (e-CERT).
- **NLP & Computer Vision:** spaCy (NER & Text Categorization), GloVe / gensim, Fuzzy Text Matching, OpenCV, PyTorch (GANs / Autoencoders in CIE Lab color space), TensorFlow.js (MoveNet).
- **Backend & Engineering:** Python (FastAPI, Pydantic), RDFLib, MySQL, REST APIs, Microservices Architecture.
- **Frontend & PWAs:** JavaScript/TypeScript, Vue.js 3 (Composition API, Pinia), CSS3 (Glassmorphism), Cytoscape.js.
- **DevOps & Infrastructure:** Docker, Jenkins (Automated CI/CD), Kubernetes, Linux (SSH / HPC).

Professional Experience & Research

Institute of Software Technology (ITS) | AI & Software Researcher

2023 – Present (Full-time Researcher 40h/wk) | 2021 – 2023 (Undergraduate Researcher 30h/wk)

- **e-CERT — Electronic Sanitary Certificate Exchange (MAPA / UN/CEFACT / OECD / CODEX):** Lead AI/NLP architect for the Brazilian Ministry of Agriculture (MAPA). Developed data normalization APIs using **spaCy (NER & textcat with GloVe embeddings)** aligned with **UNECE** vocabulary standards, currently operating live with the **European Union, China, Russia, South Korea, Australia, Chile, and the Netherlands**.
- **CONSECANA-SP & FGV / IBRE (Consultant 2024–2025):** Architecture of the Sugarcane Industry Indicator Platform with data harmonization under the SAT standard.
- **Port Community Systems (PCS / BRAEXP):** ICT architecture specification for the Santos Port Authority in partnership with ITS/USP (Scrum).

Heart Institute (InCor - HC / FMUSP) | Bioinformatics & Big Data Researcher

2018 – 2020

- **Cardiovascular Machine Learning:** Statistical predictive modeling (PLS/RCMQP in **R**) for genomic data (+300k individuals) under Prof. Dr. José Eduardo Krieger.
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Personal Ecosystem, Health AI & Projects

Personal Ecosystem, Health AI & Projects

Conception and engineering of intelligent tools, health AI systems, and automated workflows:

-  **Salvia (Full-Stack AI Clinical Assistant & GraphRAG):** Real-time clinical decision support & doctor-patient platform engineered with **Vue.js (JavaScript, WebSockets)** on the frontend and **Python (FastAPI)** on the backend, integrating vector search (ChromaDB) and neuro-symbolic medical ontologies (**SNOMED CT, ICD-10**) for evidence-based anamnesis and clinical guideline mapping.
-  **MindFlow (Journaling & Habit Tracker):** Mobile-first PWA featuring a Tinder-style Swipe Game mode, 1-touch micro-journaling, and behavioral analytics.
-  **Motion App (Calisthenics & Semantic Workouts):** Core engine in **FastAPI + RDFLib (OWL/SPARQL)** with camera pose estimation via **Vue 3 + TensorFlow.js (MoveNet)**.

Coloriz.AI | Computer Vision & Deep Learning

- Historical media chromatic restoration for 1920s archives using **PyTorch (GANs / Autoencoders in CIE Lab space)**.
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Education & Academic Qualifications

B.Sc. in Mathematics — University of São Paulo (USP)

2017 – 2023

- Thesis Research: *Machine Learning in Cardiovascular Medicine* (Advisor: Prof. Dr. José Eduardo Krieger - InCor/FMUSP).
 - Passed national **POSCOMP** exam and completed graduate-level coursework at IME-USP (KBS - Knowledge-Based Systems, Algorithm Analysis, Graph Theory) with **Grade A ("Conceito A")**.
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Languages

- **Portuguese:** Native
- **English:** Proficient (Technical Reading, Writing & Conversation)
- **French:** Intermediate